DAY 3 • LEARNING JOURNAL

Make Data-Driven Decisions by Using Looker

Day 3 Notes — Exploring Data, Building Visualizations, and Sharing Insights with Looker & Looker Studio

Looker Basics | Dimensions & Measures | Visualizations | Looker Studio | Sharing Data

Compiled from personal study notes • Google Skills: "Introduction to Data Analytics on Google Cloud"

What Is Looker?

Looker is a powerful tool that helps you:

- **Access and review** the data that your company collects.
- **Answer data questions in real time**, instead of waiting on a report to be built for you.
- **Stay up to date** with the status of your business, and use data to drive decision-making.

Analogy: If BigQuery is the engine that stores and calculates your data, Looker is like the dashboard of a car. You do not need to understand the engine to know your speed, fuel level, or engine temperature; the dashboard translates the raw mechanics into something you can act on instantly. Looker does the same for business data.

The common data analysis process in Looker

Whenever you sit down to answer a business question using Looker, you generally follow the same four steps:

1. **Identify the data questions you want to answer.** For example: "Which product category sold the most last month?"
2. **Identify the data required** to answer your question — which tables, fields, and time ranges are relevant.
3. **Analyze the data, or explore it.** You combine dimensions and measures inside Looker to find the answers.
4. **Interpret the results** and decide what action to take based on what you found.

Why this order matters: Skipping straight to "explore the data" without first writing down your actual question is the most common way analysts waste time. Starting with a clear question keeps you focused on the fields that matter.

In this module, you will practise three specific skills:

Manipulate a Looker Explore | Create the right visualization | Share visualizations with others

Use Looker to Analyze and Chart Data

What is an Explore?

In Looker, an **Explore** is a report-builder interface, and a portal to ask questions using your data. Explores are curated by Looker developers in a proprietary templating language called **LookML**, so as an analyst, you do not need to write any code to use one — someone has already built the structure for you.

Analogy: Think of an Explore like a restaurant menu that a chef (the LookML developer) has already designed. You do not need to know how each dish is cooked; you simply pick from the menu of available fields and combine them to get the meal (the answer) you want.

For example, an Explore might contain e-commerce data. The left-side panel displays the **field picker**, where fields are bundled into groups and views. There are two primary types of fields:

Dimensions

Dimensions are attributes of your data. Each column in a table is typically a dimension — for example, the date an order was placed, the sale price, or the shipping status. In Looker, all of these columns are dimensions.

Measures

Measures are calculations performed across multiple rows of data — for example, a count, sum, or average. You typically first group data by a dimension, then add a measure on top of it.

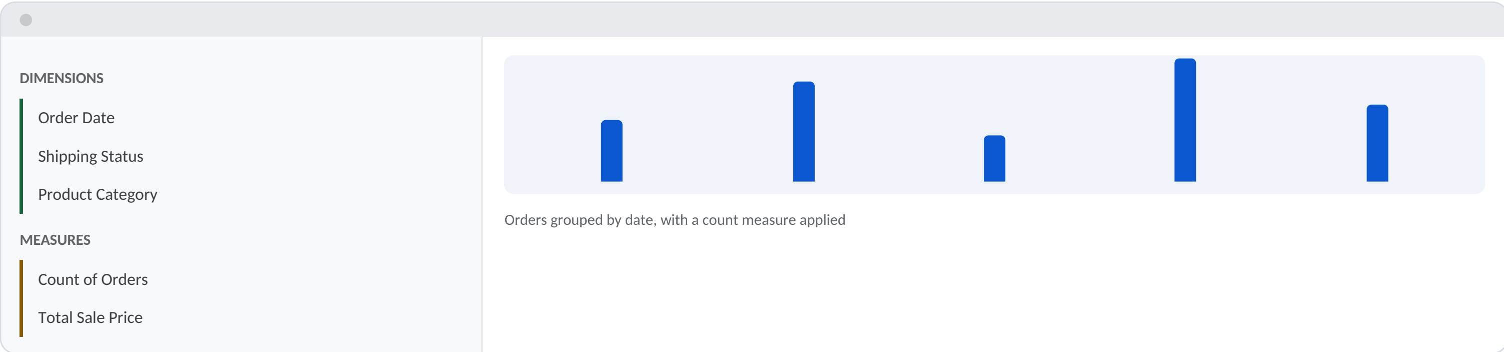
Here is a simple example: if you group an orders table by the **order date** dimension, and then add a **count** measure, Looker will show you how many orders were placed on each date.

Order Date	Sale Price	Shipping Status	Count (measure)
12 July 2026	₹1,299	Delivered	1
12 July 2026	₹799	Shipped	1
13 July 2026	₹2,499	Delivered	1

Dimension (an attribute) Measure (a calculation)

In simple words: Dimensions describe "what" something is (a date, a status, a category). Measures describe "how much" or "how many" of something there is. You will almost always need to combine dimensions and measures together to answer a real data question.

Below is a simplified, conceptual illustration of what a Looker Explore interface generally looks like:



Conceptual mockup of a Looker Explore — illustrative only.

Making Visualizations in Looker

Filters

Filters **narrow the results returned**, based on specific criteria you set — for example, showing only orders placed in the last 30 days. Importantly, filters **do not delete anything** from the underlying database; they simply change what is displayed to you. You can filter both dimensions and measures.

Pivots

Pivots **create a matrix of your data**. When you pivot on a dimension, each unique possible value of that dimension becomes its own column header, and any measures are then repeated under each column header.

Analogy: Imagine you have a list of sales with a "Region" column and a "Revenue" column. Instead of one long list, pivoting on Region turns each region (North, South, East, West) into its own column, so you can compare their revenue side-by-side at a glance — similar to a pivot table in Excel.

Looks and Dashboards

- You can save your findings to a **dashboard**, or as a **Look**.
- A **Look** is a single, saved report or visualization.
- Looks can be added to a dashboard.
- A **dashboard** is a series of saved reports, all on one page. It presents the answers to multiple data questions, and is often where data explorers store their findings.

- **Explores, Looks, and dashboards** can all be shared with others.

Explore	Look	Dashboard
Ask a question	Save one answer	Collect many Looks

Hands-on Lab: Looker Data Explorer Qwik Start

This lab is a first hands-on walkthrough of the Looker Explore interface. In this lab, you will:

1. **Use the Explore interface** to access data already published by LookML developers.
2. **Work with dimensions and measures** to query and select data.
3. **Select the appropriate visualization type** to best display your data.
4. **Save an Explore query to a dashboard**, so it can be revisited and shared later.

Visualize Data Using Looker Studio

Looker

Focused on **analyzing data and creating reports**, with a strong connection to the underlying data model (LookML).

Looker Studio

Focused on **usability and data visualization** — a more lightweight, drag-and-drop tool for building reports quickly.

Features of Looker Studio

- Easy to **share and collaborate** on reports.
- **Publish reports to the web**, and embed reports inside other applications.
- A **drag-and-drop interface** for building reports without writing code.
- Various report types and dashboards, including **charts, graphs, and tables**.
- Add **filters** and drill down into your data.
- **Real-time collaboration** on reports and dashboards, similar to editing a Google Doc together.
- **Security features** that let you control who has access to your data.

Connecting Looker and Looker Studio

- Looker and Looker Studio can be integrated using the **Looker connector for Looker Studio**.
- This integration lets you connect to any Google Cloud-hosted Looker instance, and create reports and dashboards using Looker data.

To use the Looker connector, follow these steps:

1. First, **enable the integration** in your Looker instance.
2. **Select the Looker connector** as your data source in Looker Studio.
3. **Enter the URL** of your Looker instance, and select an Explore to connect to.
4. **Start creating your report** using live Looker data.

Choosing the Right Visualization

To create a situation-appropriate visualization in a Looker Studio report, you need to consider three questions:

1. **What type of data are you working with?** (numbers over time, categories, geographic data, and so on)
2. **What insights are you trying to communicate?** (a trend, a comparison, a proportion, a distribution)
3. **Who is your audience?** (executives may want a single clear number; analysts may want a detailed breakdown)

Looker Studio visualization options

Bar Chart	Line Chart	Area Chart	Pie Chart
Donut Chart	Scatter Plot	Heatmap	More options

Quick guide: Use a **bar or line chart** for trends and comparisons over time. Use a **pie or donut chart** for proportions of a whole (but only with a few categories). Use a **scatter plot** to spot relationships between two measures. Use a **heatmap** to spot patterns across two dimensions at once.

Looker vs. Looker Studio: Which One to Use?

Both tools help you work with data, but they are built for different needs. Here is a side-by-side comparison based on today's notes:

Looker	Looker Studio
Drawback: Stores data in the cloud, which is potentially vulnerable to security breaches — though it offers encryption and access control to help protect it.	**Drawback:** Overall a less powerful platform than Looker; does not support advanced data modeling or machine learning.
Drawback: A paid platform, so cost can be a factor for some organizations.	**Drawback:** Does not integrate with as many data sources as Looker, which can make it harder to pull in data from various places.
Drawback: A relatively complex platform to learn.	**Benefit:** Very easy to use, with a simple drag-and-drop interface.
Benefit: Powerful and scalable — a strong choice for organizations that need serious, large-scale data sharing.	**Benefit:** Flexible, with various features to customize how data is shared, plus its own security features like encryption and access control.

In simple words: Looker is a good choice for organizations that need a powerful and scalable platform for data sharing. Looker Studio is a good choice for organizations that are simply in search of an easy, user-friendly platform to get reports out quickly.

Sharing Data in Looker & Looker Studio

Sharing data from Looker is often referred to as a **"data delivery."** From the settings menu, you can share your data-driven insights in a few different ways.

Sharing in Looker

- First, save your work as a report called a **Look**, or as a **dashboard** displaying several reports.

- Alternatively, to export data on a one-time basis, you can **download or send** it.
- **Downloading** extracts the data to your own computer. You can download it as a CSV, ZIP of CSVs, PDF, or PNG.
- **Sending** lets you deliver the data somewhere else — the most popular option is email.
- To send data more than once, create a **schedule** and define how often the report is sent to others.
- You can also click Share, then copy the URL and send it directly to people who need to see it.

CSV | ZIP of CSVs | PDF | PNG

Sharing in Looker Studio

- Looker Studio is built on top of **Google Drive**, so you share reports and data sources the same way you would share a Google Doc, Sheet, or Slide.
- Users are given either **edit** or **view** permission on shared data.
- You can **schedule a delivery**, get a shareable link to the report, or download it directly.

> **Very similar to Google Drive:** Since Looker Studio is built on Google Drive, if you already know how to share a Google Doc or Sheet with a colleague, you already know most of how sharing works in Looker Studio.

Sharing reports and visualizations — the bigger picture

- Data can be shared from **both Looker and Looker Studio**.
- **Sharing in Looker** provides a centralized repository for data, which ensures that data is accurate, consistent, and easy to find.
- Looker is also scalable, and can be integrated with other data sources and apps.

Hands-on Lab: Looker Studio Qwik Start

This lab introduces the Looker Studio interface. In this lab, you will:

✓ Step 1

Create a report by pulling in a public dataset from BigQuery directly into Looker Studio.

✓ Step 2

Add a chart and style the report, applying what you learned about choosing the right visualization.

> **Bringing it all together:** Today's journey went from asking a plain business question, to exploring data with dimensions and measures in Looker, to choosing the right chart for the story, and finally to sharing that story with the people who need it. This is the same loop that professional data analysts repeat every day.